WORKSHEET 4.7

Indeterminate Forms and L'Hôpital's Rule

Unit 4: Derivatives in Context

Key ideas

- If $\dfrac{f(x)}{g(x)}$ has the form $\dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$, then $\lim \dfrac{f(x)}{g(x)} = \lim \dfrac{f'(x)}{g'(x)}$, provided the second limit exists.
- Differentiate numerator and denominator **separately**. This is not the quotient rule.
- **Check the form first**, and say so: state that both limits are 0 (or both infinite) before applying the rule.
- If the result is again indeterminate, apply the rule again.

Worked examples

Example 1 · The form 0/0

Evaluate $\displaystyle\lim_{x\to 0} \frac{\sin(4x)}{3x}$.

Solution. Both $\sin(4x)$ and $3x$ tend to 0. By L'Hôpital's Rule the limit is $\displaystyle\lim_{x\to 0} \frac{4\cos(4x)}{3}$.

$\dfrac{4}{3}$.

Example 2 · The form infinity over infinity

Evaluate $\displaystyle\lim_{x\to \infty} \frac{\ln x}{x}$.

Solution. Both tend to infinity. The rule gives $\displaystyle\lim_{x\to \infty} \frac{1/x}{1}$.

0.

Example 3 · Applying the rule twice

Evaluate $\displaystyle\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$.

Solution. Form $\frac{0}{0}$: $\displaystyle\lim_{x\to 0} \frac{e^x - 1}{2x}$, again $\frac{0}{0}$: $\displaystyle\lim_{x\to 0} \frac{e^x}{2}$.

$\dfrac{1}{2}$.

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Practice

CORE

1. Evaluate $\lim_{x\to 2} \dfrac{x^2-4}{x-2}$ using L'Hôpital's Rule.

2. Evaluate $\lim_{x\to 0} \dfrac{1-\cos x}{x}$.

3. Evaluate $\lim_{x\to 1} \dfrac{x^5-1}{x-1}$.

4. Can L'Hôpital's Rule be used for $\lim_{x\to 0} \dfrac{x+1}{x+3}$? Find the limit.

EXAM LEVEL

5. Evaluate $\lim_{x\to 0} \dfrac{\tan(3x)}{\sin(2x)}$.

6. Evaluate $\lim_{x\to \infty} \dfrac{x^2+1}{e^x}$.

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7. Evaluate $\lim_{x\to 0} \dfrac{\sin x - x}{x^2}$.

8. A differentiable function f has $f(2) = 0$ and $f'(2) = 3$, and f' is continuous. Find $\lim_{x\to 2} \dfrac{f(x)}{x^3 - 8}$.

Question 9 · Free-response practice

Evaluate each limit. In each part, show that L'Hôpital's Rule applies.

(a) $\lim_{x\to 0} \dfrac{e^{2x} - 1}{\tan x}$

(b) $\lim_{x\to \pi/2} \dfrac{\cos x}{x - \frac{\pi}{2}}$

(c) $\lim_{x\to 0} \dfrac{g(x)}{e^x - 1}$, where g is differentiable with $g(0) = 0$ and $g'(0) = 5$, and g' is continuous.

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Worked solutions

1. Form $\frac{0}{0}$: $\lim_{x\to 2} \dfrac{2x}{1} = 4$.

2. Form $\frac{0}{0}$: $\lim_{x\to 0} \dfrac{\sin x}{1} = 0$.

3. Form $\frac{0}{0}$: $\lim_{x\to 1} 5x^4 = 5$.

4. No: the numerator tends to 1 and the denominator to 3, so the form is not indeterminate. The limit is $\dfrac{1}{3}$.

5. Form $\frac{0}{0}$: $\lim_{x\to 0} \dfrac{3\sec^2(3x)}{2\cos(2x)} = \dfrac{3}{2}$.

6. Form $\frac{\infty}{\infty}$, twice: $\lim_{x\to\infty} \dfrac{2x}{e^x} = \lim_{x\to\infty} \dfrac{2}{e^x} = 0$.

7. Form $\frac{0}{0}$, twice: $\lim_{x\to 0} \dfrac{\cos x - 1}{2x} = \lim_{x\to 0} \dfrac{-\sin x}{2} = 0$.

8. Both $f(x)$ and $x^3 - 8$ tend to 0. $\lim_{x\to 2} \dfrac{f'(x)}{3x^2} = \dfrac{3}{12} = \dfrac{1}{4}$.

9. (a) $e^{2x} - 1 \to 0$ and $\tan x \to 0$. $\lim_{x\to 0} \dfrac{2e^{2x}}{\sec^2 x} = 2$.

   (b) $\cos x \to 0$ and $x - \dfrac{\pi}{2} \to 0$. $\lim_{x\to \pi/2} \dfrac{-\sin x}{1} = -1$.

   (c) $g(x) \to g(0) = 0$ because g is continuous, and $e^x - 1 \to 0$. $\lim_{x\to 0} \dfrac{g'(x)}{e^x} = \dfrac{5}{1} = 5$.

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